

CP2a Resultant Forces



Resultant force

Learning Outcome: Understand how to represent all the forces acting on an object

TROGS

Task and time: What is a scalar? What is a vector?

5 minutes

Resources: Exercise book

Outcome: Understand the difference between a scalar and a vector and how speed, velocity, and acceleration are connected. Retrieval practice completed as extension task.

Grouping: Individual work **in silence**.

Stop Signal: I will raise my hand, and you will raise yours to show you are ready to listen.

Vector and scalar quantities

LO: To understand the difference between vector and scalar quantities.

RECAP ACTIVITY:

What are vector and scalar quantities?

Write down five things that you can remember about forces.

RETRIEVAL PRACTICE:

1. What is a force?
2. How do you measure force?

Keywords:

Force, resultant force, scalar, vector, acceleration, balanced, unbalanced

Careers

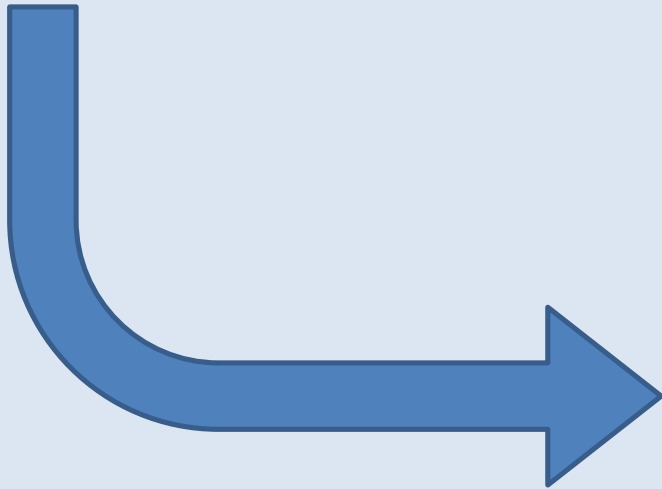
Motor racing, Sports coach
Personal trainer, Mechanic

Write down five things that you can remember about forces.



RECAP: Vectors and Scalars

A **vector** quantity has both a size and a direction. **Scalars** only have a size (magnitude).



Forces are **vector** quantities, we need to know the **direction** the force is acting as well as how big it is.

Rearrange the anagrams to work out the names of some forces:

GIEHWT

= WEIGHT

FILT

= LIFT

GARD

= DRAG

STURTH

= THRUST

SUPH

= PUSH

TINCRIOF

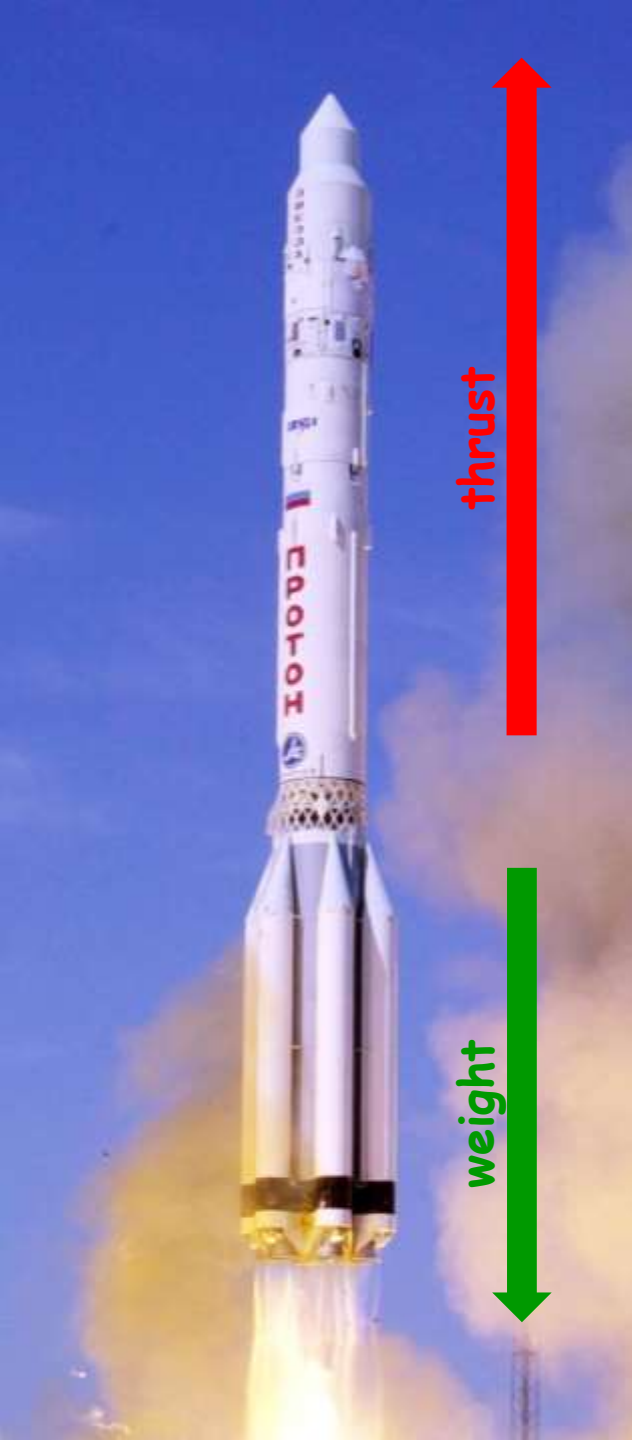
= FRICTION

STUURPTH

= UPTHrust

RIA ANETCSISRE

= AIR RESISTANCE



Representing Forces

We represent **forces** by drawing **arrows**.

The length of the arrow represents the **size** of the force.

Name that Force:

Write down the names of the forces acting on the following objects.

Resultant force

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TROGS

Task and time: Write some examples of scalar and vector quantities.
What is the connection between the speed, velocity and acceleration?

5 minutes

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Resultant force

LO: How do we represent all the forces acting on an object?

RECAP ACTIVITY:

Write some examples of scalar and vector quantities.

What is the connection between the speed, velocity and acceleration?

RETRIEVAL PRACTICE:

1. What is velocity?
2. How do you calculate velocity?

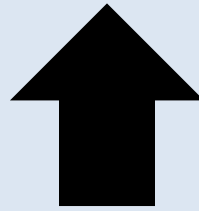
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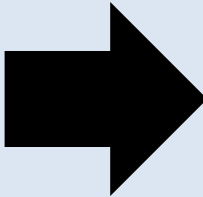
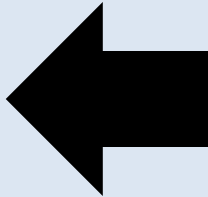
Careers

Motor racing, Sports coach
Personal trainer, Mechanic

Upwards force
from road

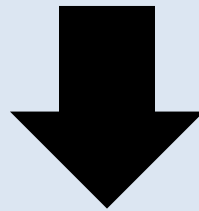


Forwards
force from
engine

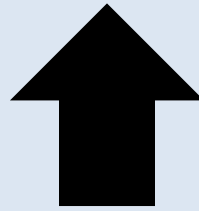


Friction and
air
resistance

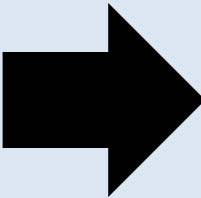
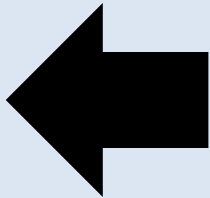
weight



Lift due to
wings

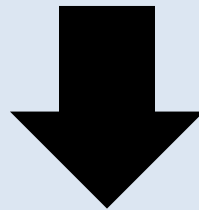


Thrust from
engines

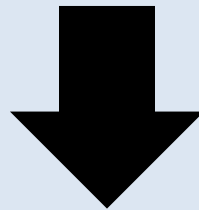
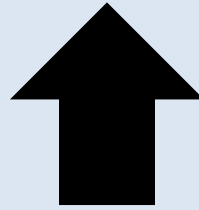


Drag (air
resistance)

weight

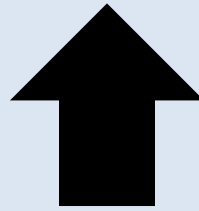


Upwards force
from rings

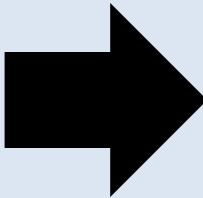
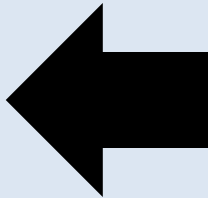


weight

upthrust

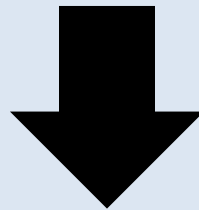


Drag (water resistance)



Forward
force from
flippers

weight



Resultant force

The **resultant** force is the
.....
overall..... force on the object.

The **thrust** is larger than
.....
weight....., so the resultant force
is **upwards**..... .

Extending: The weight is larger
than thrust, so resultant force is
downwards..... .

KEYWORDS:



thrust

overall

resultant

upwards

weight





Just Dance!

Resultant forces dance mat.

Move in the direction of the resulting force.

<https://www.youtube.com/watch?v=ZlyYFWAYrSY>



Resultant Forces

Calculating the **resultant force**:

- If they are in the same direction, you add them together.
- If they are going opposite directions you subtract the smaller one from the bigger one.



Explain the meaning of the words:

Balanced and Unbalanced

a situation in which different elements are equal or in the correct proportions

upset or disturb the equilibrium of (a situation or person's state of mind)

Resultant Forces

If the **resultant force** is 0 then the object is either:

- Stationary (not moving)
- Moving at a constant speed

We say that the forces are
BALANCED.



Resultant Forces

If the **resultant force** is NOT equal to 0 then the object is either:

- Accelerating
- Decelerating

We say that the forces are
UNBALANCED.



Examples:

- Balanced
- Unbalanced

Resultant Forces

In your book draw a diagram of **balanced** forces and unbalanced **forces**.

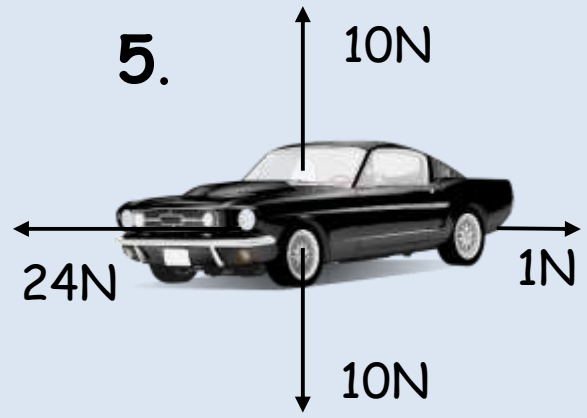
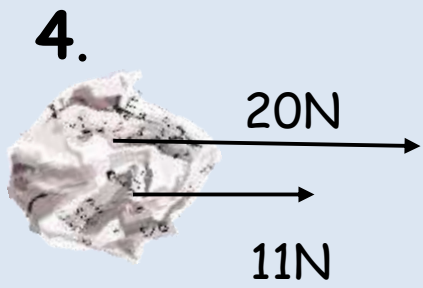
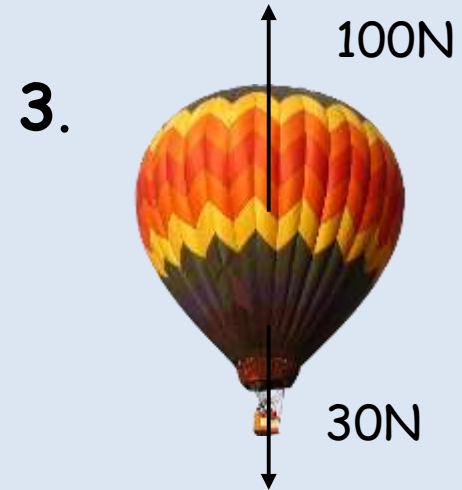
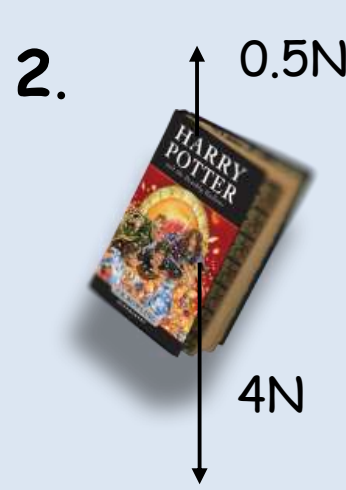
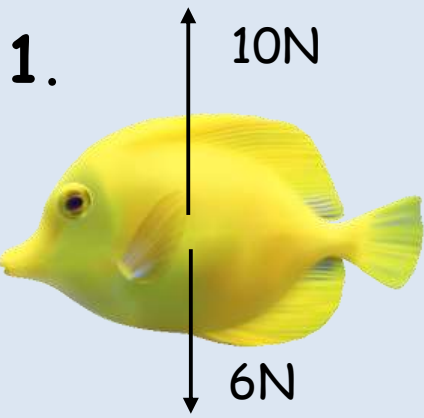
Balanced:



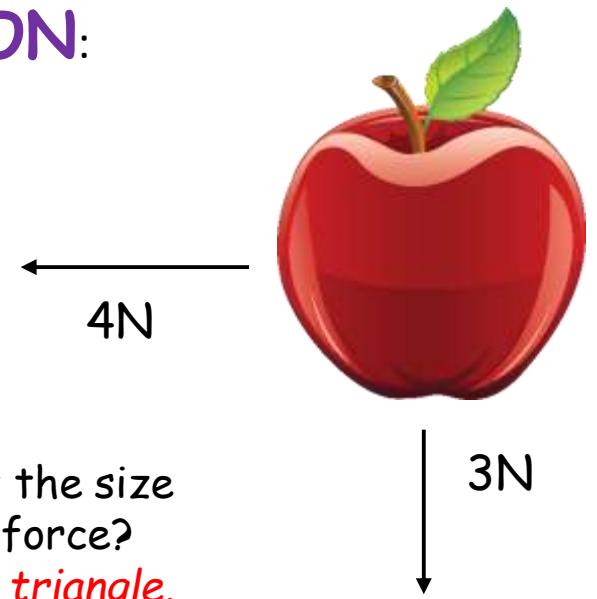
Unbalanced:



Calculate the resultant force:



EXTENSION:



Can you work out the size of this resulting force?
Think of it like a triangle.

ANSWERS:

1. 4N $\uparrow$

2. 3.5N $\downarrow$

3. 70N $\uparrow$

4. 31N $\rightarrow$

5. 23 $\leftarrow$

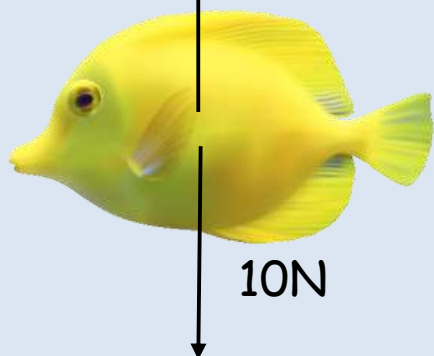
EXTENSION: 5N $\swarrow$

PEER ASSESSMENT
Mark the answers using
a **GREEN PEN**



Choose one answer for each question: a) stationary or b) constant speed or c) acceleration or d) deceleration

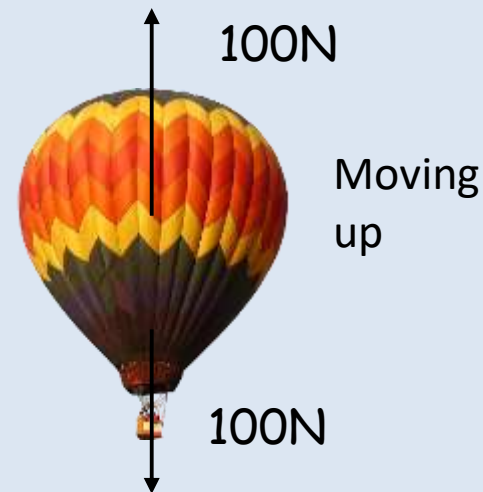
1.



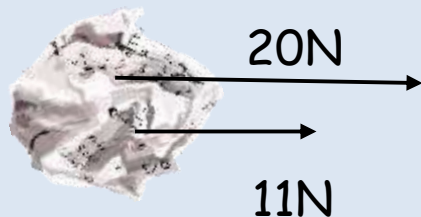
2.



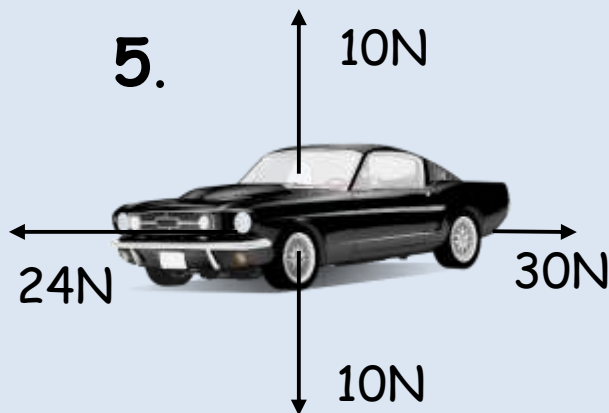
3.



4.



5.



EXTENSION:

Explain why you have chosen a) or b) or c) or d) for question 5.

ANSWERS:

1. balanced
2. acceleration $\downarrow$
3. Constant speed $\uparrow$
4. acceleration $\rightarrow$
5. deceleration $\rightarrow$

PEER ASSESSMENT
Mark the answers using
a **GREEN PEN**





Write 5 questions in your book to test someone. Try to cover all the learning outcomes from today.

Swap books and try to answer the questions now.

Learning Outcomes:

- ✓ **Understand** what are balanced and balanced forces and resultant forces.
- ✓ **Know** how to draw balanced and unbalanced forces on a diagram.
- ✓ **Describe** the diagram of balanced and unbalanced forces.
- ✓ **Use** scientific language to explain how an object will move if the forces are acting on an object are balanced or unbalanced.
- ✓ **Explain** whether forces on an object are balanced or unbalanced.